LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flow over bodies; bluff bodies and streamlined bodies, Flow pattern over flat plate, formation of eddies, the wake and its reduction.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flow over bodies; bluff bodies and streamlined bodies, Flow pattern over flat plate, formation of eddies, the wake and its reduction.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flow visualization diagrams; Safety: No physical handling of models; Instructor supervision: Monitor diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What defines a bluff body?
(b) How does flow behave over a flat plate?
(c) What is the primary difference between laminar and turbulent flow?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of bluff and streamlined bodies • Flow pattern over flat plate • Formation of eddies behind bodies • Characteristics of the wake region • Methods for wake reduction	L & Ds	PPT, IB	15 min
(b)	• Drag components on bluff bodies • Pressure drag vs friction drag • Streamlining techniques • Impact of Reynolds number • Real-world aircraft examples	L & Ds	PPT, PDF	20 min
(c)	• Boundary layer separation • Vortex shedding phenomena • Wake stability factors • Design implications for rotors • Noise generation from wakes	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how streamlining reduces wake formation.
(b) Describe the relationship between eddies and pressure drag.
(c) Identify two methods to reduce wake in helicopter design.

12. Home Assignment and Follow Up	Prepare a brief note comparing the flow characteristics of a flat plate versus a bluff body, referencing the boundary layer behavior discussed in LP 06.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Boundary layer, laminar, transition, turbulent B/L,	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Major, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min